

HARSH RAJ

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Google Scholar ◇ Github ◇ LinkedIn

EDUCATION

Northeastern University

May 2026 (expected)

M.S. in Computer Science

GPA: 3.5/4.0

Related courses: Foundations of AI, Algorithms, Human-Computer Interaction

Delhi Technological University (DTU)

June 2023

B.Tech. in Engineering Physics

GPA: 3.4/4.0

Related courses: Machine Learning, Algorithmic Game Theory, Computer Vision.

RESEARCH INTERESTS

I am interested in large-scale data processing systems and training large models at scale.

EXPERIENCE

Research Assistant, Forschungszentrum Jülich

July 2025 – Present

Remote

Cologne, Germany

- Contributing to European Union's Open Foundational Model projects, including MINERVA, OpenEuroLLM, and ELLIOT.

Research Assistant, Interpretable Neural Networks Lab

Jan 2025 – Present

On-Site

Boston, US

- Working with David Bau and Robert West on interpretability of reasoning models.

Co-Founder, Ontocord AI

Oct 2024 – Present

Remote

California, US

- Leading the training of AuroraM-2.
- Released the MixtureVitae dataset - the strongest permissibly licensed pretraining dataset to date.

Research Fellow, AI Risk and Vulnerability Alliance (ARVA)

Oct 2024 – June 2025

Remote

Washington, US

- Contributed multiple bias benchmarks to the UK AISI inspect-ai framework.
- Integrated the inspect-ai framework with the AVID database.

Applied Scientist, Vijil AI

Feb 2024 – Oct 2024

Remote

California, US

- Developed vijil-fuzzer, an LLM red-teaming framework for enterprise-grade customers.
- Integrated open-source benchmarks into the vijil-evaluation service.
- Contributed to vijil-dome, integrating guardrails into tools like Langchain with minimal latency impact.

ML Engineer, Yield Protocol

Sept 2022 – Jan 2024

Remote

Chicago, US

- Led data synthesis to improve LLM agents and built synchaev, a framework for agent data generation.
- Developed mandrill for finetuning and testing agents. Supervised 10 research interns at UIUC's Disruption Lab as part of the project.

- Contributed to the development of Cacti, a Web3 transaction chatbot.

Applied Researcher (intern), Thoucentric

Remote

May 2022 – Aug 2022

Bangalore, India

- Developed a natural language to SQL conversion system for data analytics including automatic visualization.
- Improved the previous state-of-the-art by introducing additional processing steps, boosting accuracy by 10%. Authored a white paper.

Data Scientist (intern), Attryb Tech

On-Site

June 2021 – Dec 2021

Bangalore, India

- Developed the backend for Content Studio, a tool for generating and analyzing content for blogs, product marketing, etc.
- Finetuned Pegasus and GPT-2 for multiple components of the content generation pipeline.
- Mined data from SemRush and Common Crawl for content database creation.

PUBLICATIONS

- **Harsh Raj**, Vipul Gupta, Domenic Rosati, Subhabrata Majumdar. *Improving Consistency in Large Language Models through Chain of Guidance* (TMLR'2025).
- Domenic Rosati, Giles Eddins, **Harsh Raj**, David Atanasov, Kai Williams, Subhabrata Majumdar, Janarthanan Rajendran, Frank Rudzicz, Hassan Sajjad. *Defences against Reverse Preference Attacks* (AI Security Workshop, AAI'25).
- Laura Fee Nern, **Harsh Raj**, Maurice Georgi, Yash Sharma. *On Transfer of Adversarial Robustness from Pretraining to Downstream Tasks* (NeurIPS'23).
- **Harsh Raj**, Domenic Rosati, Subhabrata Majumdar. *Measuring Reliability of Large Language Models through Semantic Consistency* (Best Paper Award, ML Safety Workshop, NeurIPS'22).
- Sinchani Chakraborty, **Harsh Raj**, Srishti Gureja, Tanmay Jain, Atif Hassan, Sayantan Basu. *Evaluating the Robustness of Biomedical Concept Normalization* (Transfer Learning Workshop, NeurIPS'22).
- **Harsh Raj**, Ashutosh Pandey, Shaurya Kumar, Kavinder Singh, Nihal Kumar, Anil Singh Parihar. *Decoding Percepts in Vision Language Navigation: Is it about better features or more data?* (ACM TIST).
- Manu Joseph, **Harsh Raj**. *GANDALF: Gated Adaptive Network for Deep Automated Learning of Features* (under review, TMLR).
- Manu Joseph, **Harsh Raj**, Abhinav Yadav, Aaryamann Sharma. *AskYourDB: An end-to-end system for querying and visualizing relational databases using natural language*.
- **Harsh Raj**, Aakansha Gupta, Rahul Katarya. *Extract It! Product Category Extraction by Transfer Learning* (CICT'22).

PROJECTS AND LIBRARIES

- Mike Meryll, Alex Shaw, **Harsh Raj**, .., Ludwig Schmidt, Andy Kowsinski. *terminal_bench* (2025).
- **Harsh Raj**, Domenic Rosati, Sundar Sripada, Oliver Savolainen. *SafeRL-Gym* (2025).
- Leon Derczynski, Erick Galinkin, Subho Majumdar, .., **Harsh Raj**, .., Erwan Roussel, Eric Hacker. *garak* (2024).

AWARDS, GRANTS AND HONORS

SWISS AI Grant: Although not PI, contributed significantly to shaping and writing the grant proposal. The focus of this grant is to acquire infrastructure for the project "*From training a model to raising a model*". The National Supercomputing Center (CSCS) granted access to 200k GPU hours on the Alps supercomputer and 61k CHF in monetary funds. 2025

EuroHPC: As part of the OpenEuroLLM project, contributed significantly to acquiring 4M GPU hours on the LEONARDO and JUPITER supercomputers for large-scale language model training. 2024

Vision and Language Navigation: Secured 3rd position on the R2R benchmark 2022

ML Safety Challenge: Best Paper Award with a \$5000 prize 2022

Codeforces ML Round: Rank 7 in the Raif ML Round 1 2021

SKILLS/HOBBIES

Programming Languages	Python, C/C++, MATLAB, HTML
Machine Learning Tools	Pytorch, Tensorflow, Sklearn, Pandas, Numpy
Hobbies	reading and hiking